



Earth System

CHAPTER 2

Earth System Science

Chapter 2: THE COMPONENTS OF THE EARTH SYSTEM



Objective

- *To Define atmosphere*
- *To Know The Composition Of The Atmosphere*
- *To identify permanent and variable gases*
- *To see the Vertical Structure of the Atmosphere*

The Atmosphere

The Atmosphere



- The earth's atmosphere as viewed from space.
- The atmosphere is the thin **blue region** along the edge of the earth

- It is this thin blanket of air that constantly shields the surface and its inhabitants from the sun's dangerous ultraviolet radiant energy
- 99 % of the atmosphere lies within 30 km height

- Our *atmosphere* is a delicate life-giving blanket of air that surrounds the earth.
- In one way or another, it influences everything we see and hear — it is intimately connected to our lives.
- Air is with us from birth, and we cannot detach ourselves from its presence.

- We may be able to survive without food for **a few weeks**, or without water **for a few days**, but, without our atmosphere, we would not survive more than **a few minutes**.
- Just as fish are confined to an environment of water, so we are confined to an ocean of air. Anywhere we go, it must go with us.

- The earth without an atmosphere would have no lakes or oceans. There would be no sounds, no clouds, no red sunsets.
- The beautiful pageantry of the sky would be absent.
- It would be unimaginably cold at night and unbearably hot during the day.
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- Even though air is tasteless, odorless, and (most of the time) invisible, it protects us from the scorching rays of the sun and provides us with a mixture of gases that allows life to flourish.

Composition of the Atmosphere

COMPOSITION OF THE ATMOSPHERE

- Notice that **nitrogen** (N₂) occupies about 78 percent and **oxygen** (O₂) about 21 percent of the total volume of dry air.
- If all the other gases are removed, these percentages for nitrogen and oxygen hold fairly constant up to an elevation of about 80 km (50 mi).

- At the surface, there is a balance between destruction (output) and production (input) of these gases. For example, nitrogen is removed from the atmosphere primarily by biological processes that involve soil bacteria.
- In addition, nitrogen is taken from the air by tiny ocean-dwelling plankton that convert it into nutrients that help fortify the ocean's food chain. It is returned to the atmosphere mainly through the decaying of plant and animal matter.

- Oxygen, on the other hand, is **removed** from the atmosphere when organic matter decays and when oxygen combines with other substances, producing oxides.
- It is also taken from the atmosphere **during breathing**, as the lungs take in oxygen and release carbon dioxide (CO₂).
- The addition of oxygen to the atmosphere occurs during **photosynthesis**, as plants, in the presence of sunlight, combine carbon dioxide and water to produce sugar and oxygen.

- The concentration of the invisible gas **water vapor** (H_2O), however, varies greatly from place to place, and from time to time.
- Close to the surface in warm, steamy, tropical locations, water vapor may account for up to 4 percent of the atmospheric gases, whereas in colder arctic areas, its concentration may dwindle to a mere fraction of a percent.

- **water vapor** (H₂O), releases large amounts of heat — called *latent heat*.
- Latent heat is an important source of atmospheric energy, especially for storms, such as thunderstorms and hurricanes.
- water vapor is a potent *greenhouse gas*. *How?*

This is because the amount of water vapor in the atmosphere is influenced by temperature. Warmer temperatures lead to increased evaporation and, consequently, higher levels of water vapor in the atmosphere.

This, in turn, can enhance the greenhouse effect, contributing to a positive feedback loop.

- Human activities, such as the burning of fossil fuels, release greenhouse gases like carbon dioxide (CO₂) into the atmosphere, leading to an enhanced greenhouse effect.
- While *water vapor is a key player in the natural greenhouse effect*, it is the human-induced increase in other greenhouse gases that is primarily responsible for the current trends in global warming.

▼ **TABLE 1** · Composition of the Atmosphere near the Earth's Surface

PERMANENT GASES			VARIABLE GASES			
Gas	Symbol	Percent (by Volume) Dry Air	Gas (and Particles)	Symbol	Percent (by Volume)	Parts per Million (ppm)*
Nitrogen	N ₂	78.08	Water vapor	H ₂ O	0 to 4	
Oxygen	O ₂	20.95	Carbon dioxide	CO ₂	0.038	385*
Argon	Ar	0.93	Methane	CH ₄	0.00017	1.7
Neon	Ne	0.0018	Nitrous oxide	N ₂ O	0.00003	0.3
Helium	He	0.0005	Ozone	O ₃	0.000004	0.04†
Hydrogen	H ₂	0.00006	Particles (dust, soot, etc.)		0.000001	0.01–0.15
Xenon	Xe	0.000009	Chlorofluorocarbons (CFCs)		0.00000002	0.0002

*For CO₂, 385 parts per million means that out of every million air molecules, 385 are CO₂ molecules.

†Stratospheric values at altitudes between 11 km and 50 km are about 5 to 12 ppm.

▼ **TABLE I** Data on Planets and the Sun

	DIAMETER	AVERAGE DISTANCE FROM SUN	AVERAGE SURFACE TEMPERATURE		MAIN ATMOSPHERIC COMPONENTS
	Kilometers	Millions of Kilometers	°C	°F	
Sun	$1,392 \times 10^3$		5,800	10,500	—
Mercury	4,880	58	260*	500	—
Venus	12,112	108	480	900	CO ₂
Earth	12,742	150	15	59	N ₂ , O ₂
Mars	6,800	228	−60	−76	CO ₂
Jupiter	143,000	778	−110	−166	H ₂ , He
Saturn	121,000	1,427	−190	−310	H ₂ , He
Uranus	51,800	2,869	−215	−355	H ₂ , CH ₄
Neptune	49,000	4,498	−225	−373	N ₂ , CH ₄
Pluto	3,100	5,900	−235	−391	CH ₄
*Sunlit side.					

Vertical Structure of the Atmosphere

Vertical Structure of the Atmosphere

Revising about Air Density and Air Pressure

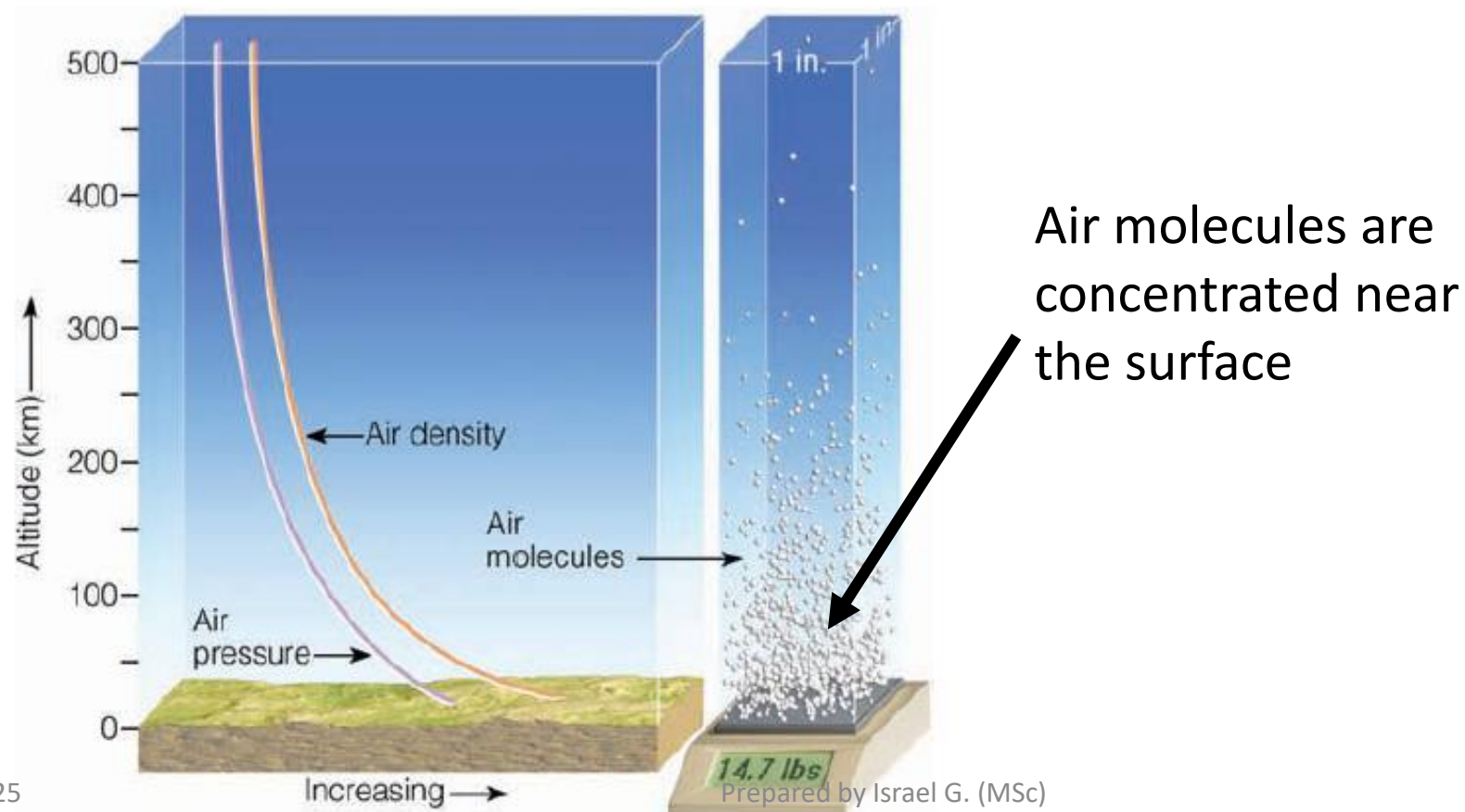
- ✓ The **density of air** is determined by the masses of atoms and molecules and the amount of space between them.
- ✓ It tells us how much matter is in a given space (that is, volume)

$$\text{Density} = \frac{\text{mass}}{\text{volume}}$$

- ✓ Air density is greatest at the surface and decreases as we move up into the atmosphere. Why?

Because there are more molecules within the same size volume of air near the earth's surface than at higher levels

What about the rate of decrease? Is it the same up until the top?
decreases rapidly at first, then more slowly as we move farther away from the surface: because air near the surface is compressed

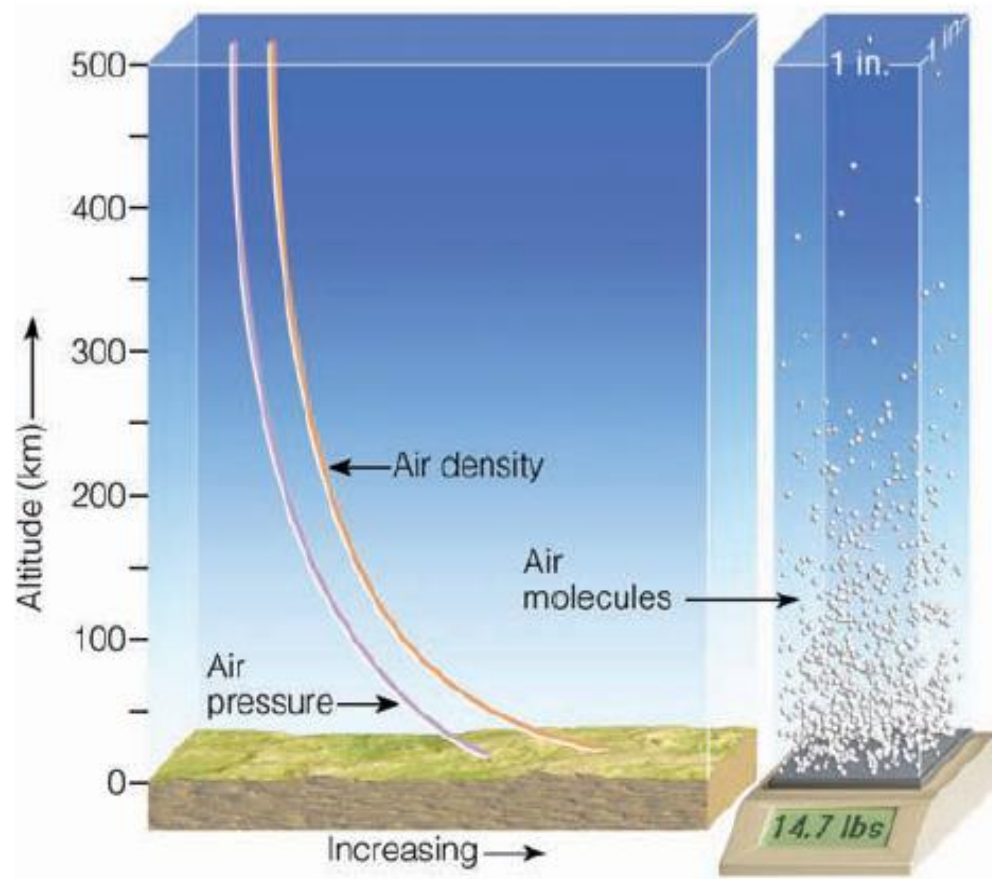


- ✓ Air molecules are in constant motion giving a tiny push every time they bounce against an object.
- ✓ This small force (push) divided by the area on which it pushes is called **pressure**; thus

$$Pressure = \frac{Force}{Area}$$

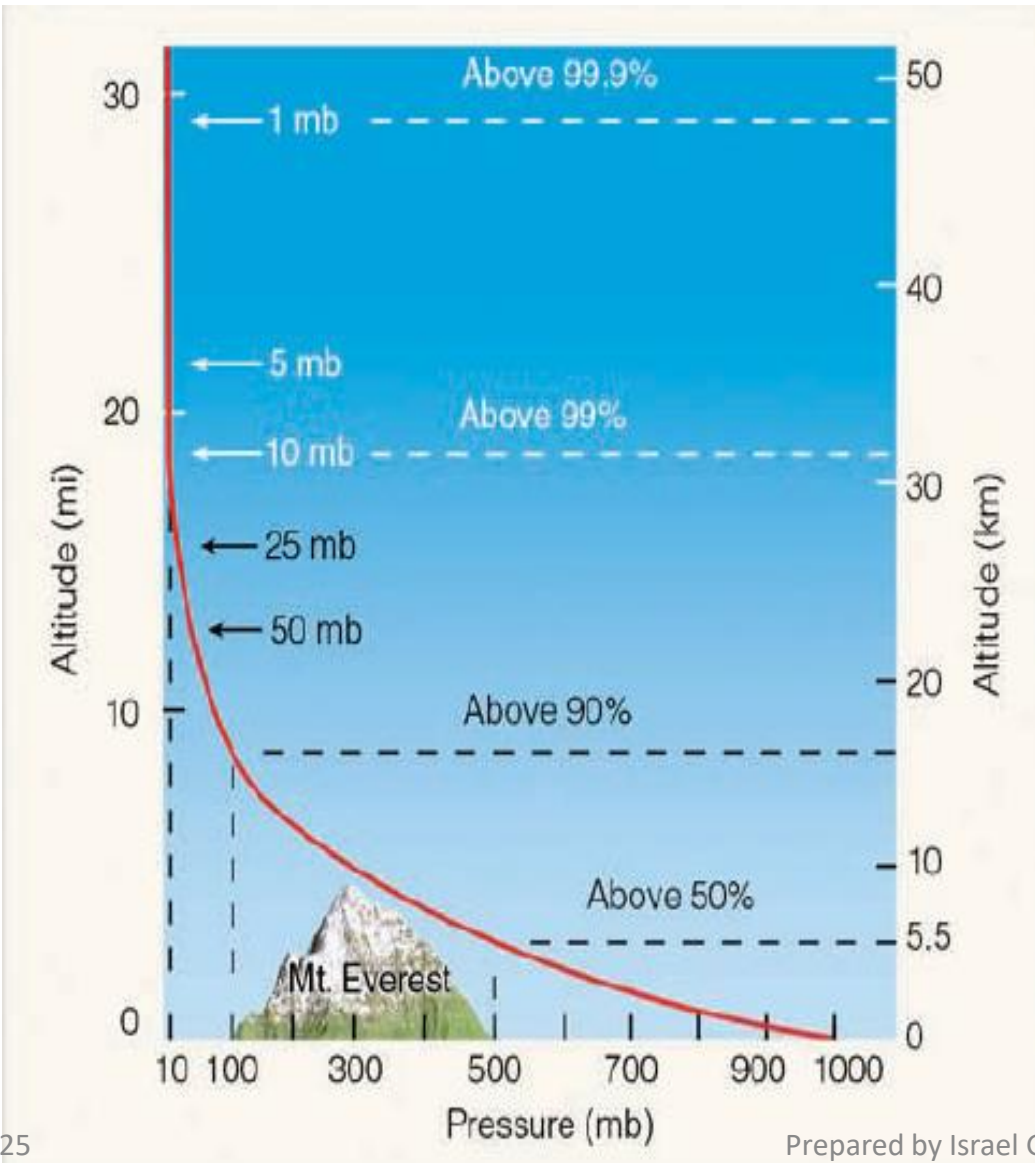
- ✓ If we weigh a column of air 1 square inch in cross section, extending from the sea level to the “top” of the atmosphere, it would weigh nearly **14.7 pounds**.
- ✓ Thus, normal atmospheric pressure near sea level is close to 14.7 pounds per square inch (**1012.83 mb or hpa**).

✓ *air pressure decreases with increasing altitude*



✓ At sea level the normal atmospheric pressure is 1012 mb or hp

A closer look at the rate of decrease in air pressure with altitude



- ✓ Near sea level, atmospheric pressure is usually close to 1000 mb.
- ✓ Just above sea level, atmospheric pressure decreases by about 10 mb for every 100 meters (m) increase in elevation.

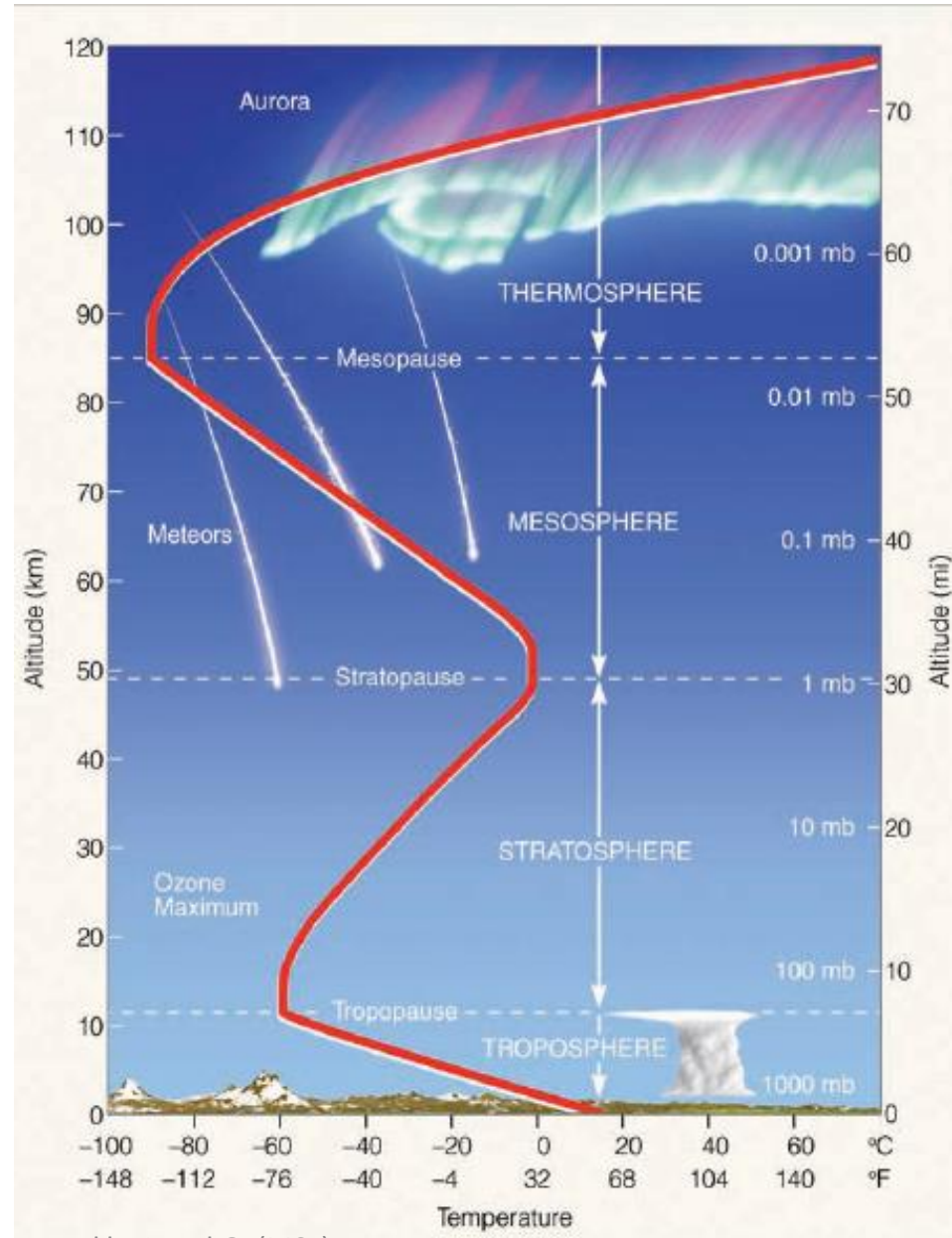
- ✓ As with air density, air pressure decreases fast at lower level
- ✓ at an altitude of only 5.5 km, the air pressure is about 500 mb, or half of the sea-level pressure.
- ✓ At an altitude approaching 50 km, the air pressure is about 1 mb

Layers of the Atmosphere

✓ *Air temperature* is the degree of hotness or coldness of the air and it is also a measure of the average speed of the air molecules.

✓ What is Earth's average surface temperature?

15°C



- ✓ There are four layers of Atmosphere:
- Troposphere
 - Stratosphere
 - Mesosphere
 - Thermosphere

Troposphere

- ✓ The region of the atmosphere from the surface up to about 11 km is kept well stirred by rising and descending air currents.
- ✓ This region from the surface up to the height where temperature stops decreasing with height is called **Troposphere**
- ✓ Just above 11 km the air temperature normally stops decreasing with height. What about the lapse rate?
- ✓ This region, where, on average, the air temperature remains constant with height is called **tropopause**

- ✓ air temperature normally decreases from the earth's surface up to an altitude of about 11 km. Why?

Because sunlight warms the earth's surface, and the surface, in turn, warms the air above it.

- ✓ The rate at which the air temperature decreases with height is called the temperature **lapse rate**.
- ✓ The *average lapse rate* in this region is about **6.5°C for every 1000 m rise in elevation**.

- ✓ Most of Earth's weather happens here, and almost all clouds that are generated by weather are found here
- ✓ also contains about 99 percent of all water vapor and aerosols (minute solid or liquid particles suspended in the atmosphere).
- ✓ Most aviation takes place here, including in the transition region between the troposphere and the stratosphere.

Stratosphere

- ✓ The next atmospheric layer that lay above tropopause is called **stratosphere**.
- ✓ Located between approximately 12 and 50 kilometers (7.5 and 31 miles) above Earth's surface,
- ✓ In the stratosphere, the air temperature begins to increase with height
- ✓ A condition in which air temperature increases with height is known as a **temperature inversion**.

- ✓ The stratosphere is perhaps best known as home to *Earth's ozone layer*, which protects us from the Sun's harmful *ultraviolet radiation*.
- ✓ Because of that UV radiation, the higher up you go into the stratosphere, the warmer temperatures become.
- ✓ The stratosphere is nearly cloud- and weather-free, but polar stratospheric clouds are sometimes present in its lowest, coldest altitudes.
- ✓ It's also the highest part of the atmosphere that jet planes can reach.

Mesosphere

- ✓ Atmospheric layer above the stratosphere is the **mesosphere** and the boundary separating these two layers is called **Stratopause**.
- ✓ Located between about 50 and 85 kilometers (31 and 45 miles) above Earth's surface, the mesosphere gets progressively colder with altitude.
- ✓ The air at this level is extremely thin and the atmospheric pressure is quite low, averaging about 1 mb
- ✓ Absence of ozone is one of the reason for decreasing temperature with elevation in this layer.

- In fact, the top of this layer is the coldest place found within the Earth system, with an average temperature of about minus 90 degrees Celsius.

Thermosphere

- ✓ The “hot layer” above the mesosphere is the **Thermosphere**.
- ✓ Located between about 85 and 600 kilometers (50 and 400 miles) above Earth’s surface is the thermosphere, whose lowest part contains the ionosphere.
- ✓ The boundary that separates the lower, colder mesosphere from the warmer thermosphere is the ***mesopause***.
- ✓ Because there are relatively few atoms and molecules in the thermosphere, the absorption of a small amount of energetic solar energy can cause a large increase in air temperature.

Exosphere

- Located between about 600 and 10,000 kilometers (440 and 6,200 miles) above Earth's surface, the exosphere is the highest layer of Earth's atmosphere and, at its top, merges with the solar wind.
- Molecules found here are of extremely low density, so this layer doesn't behave like a gas, and particles here escape into space.
- There's no weather at all in the Exosphere.
- Most Earth satellites orbit in the exosphere.

The Radiosonde

- The vertical distribution of temperature, pressure, and humidity up to an altitude of about 30 km can be obtained with an instrument called a *radiosonde*.
- The radiosonde is a small, lightweight box equipped with weather instruments and a radio transmitter.
- It is attached to a cord that has a parachute and a gas-filled balloon tied tightly at the end (see Fig.)



Earth System Science

Chapter 2: THE COMPONENTS OF THE EARTH SYSTEM



Objective

- *To Know General Circulation of the Atmosphere*
- *To Explain Ozone-oxygen Cycle*
- *To See Biogeochemical reactions in the atmosphere*
- *To Identify Ozone Depletion and Ozone Hole*

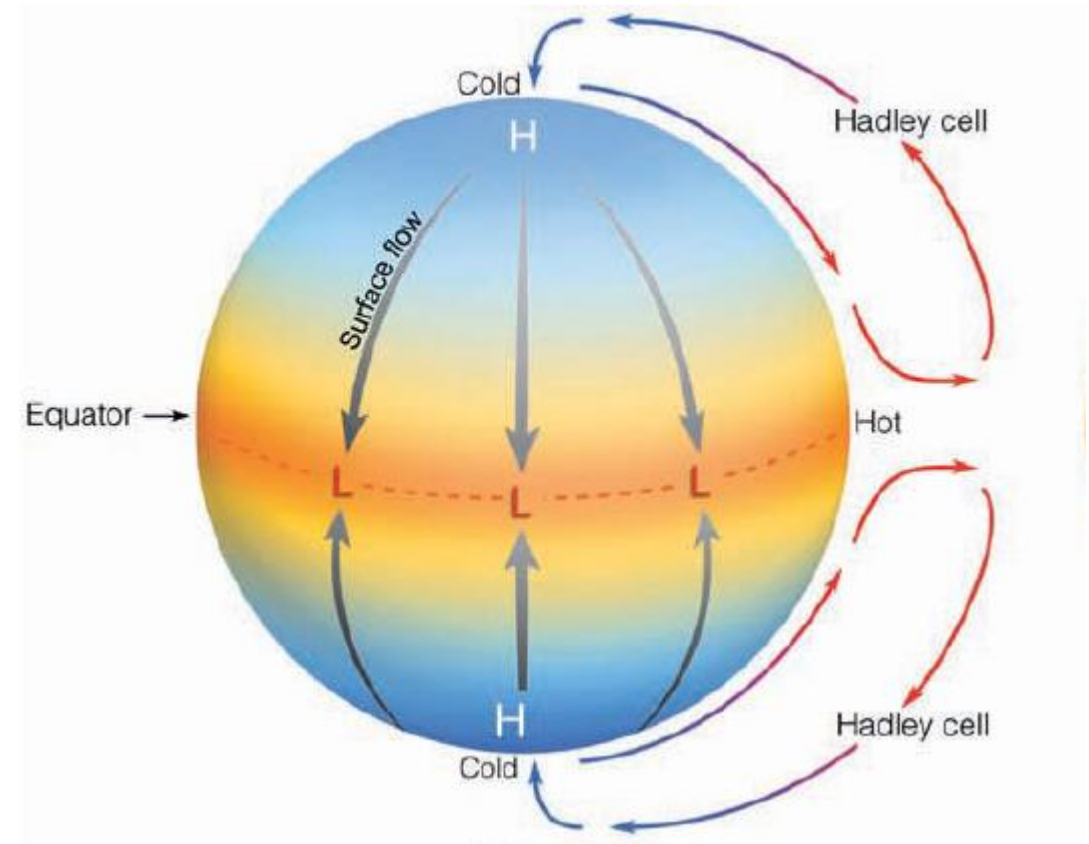
General Circulation of the Atmosphere

- The underlying cause of the general circulation is the *unequal heating of the earth's surface*.

SINGLE-CELL MODEL in which we assume that:

1. The earth's surface is uniformly covered with water.
2. The sun is always directly over the equator.
3. The earth does not rotate.

Excessive at the equator produces a broad region of surface low pressure, while at the poles excessive cooling creates a region of surface high pressure.

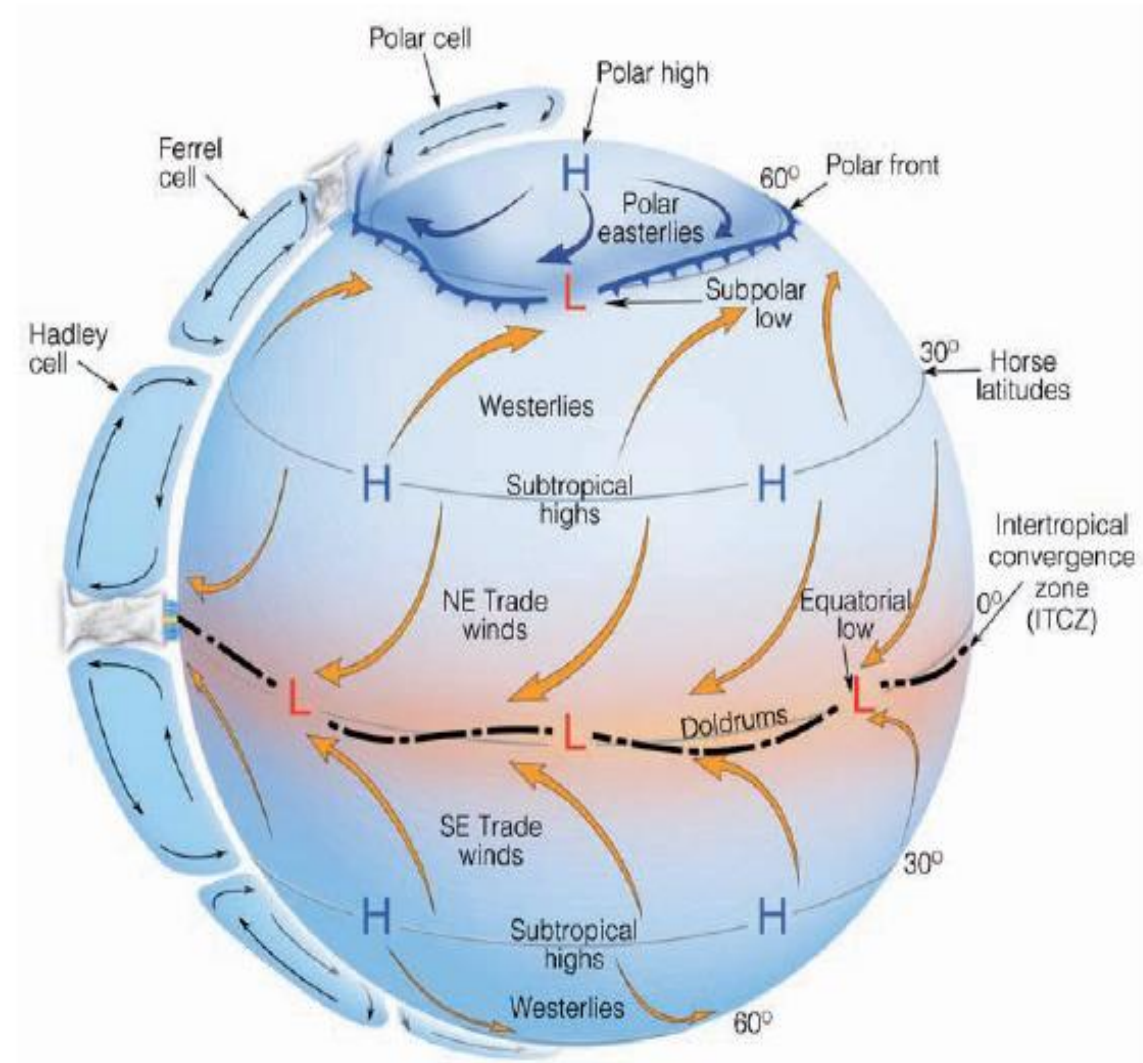


- ✓ *The Coriolis force deflects this poleward flow toward the right in the Northern Hemisphere and to the left in the Southern Hemisphere*
- ✓ As it moves poleward it cools constantly by giving up infrared radiation
- ✓ it also begins to converge, especially as it approaches the middle latitudes (30°N/S) and starts to descend.

THREE-CELL MODEL *this let us take* that the earth is covered with water and that the sun is always directly above the equator. But let us allow the earth to spin or rotate.

- ✓ The warm air, which is warmed by the excessive equatorial heating, at the equator rises.
- ✓ When it reaches the tropopause, which acts like a barrier, it is caused to move laterally toward the poles.

- ✓ **Some of** the descended air moves as surface air back toward the equator under the influence of Coriolis force.
- ✓ Some air moves toward the poles and deflects toward the east
- ✓ As it moves poleward, it encounters cold air moving down, forming **polar front**.

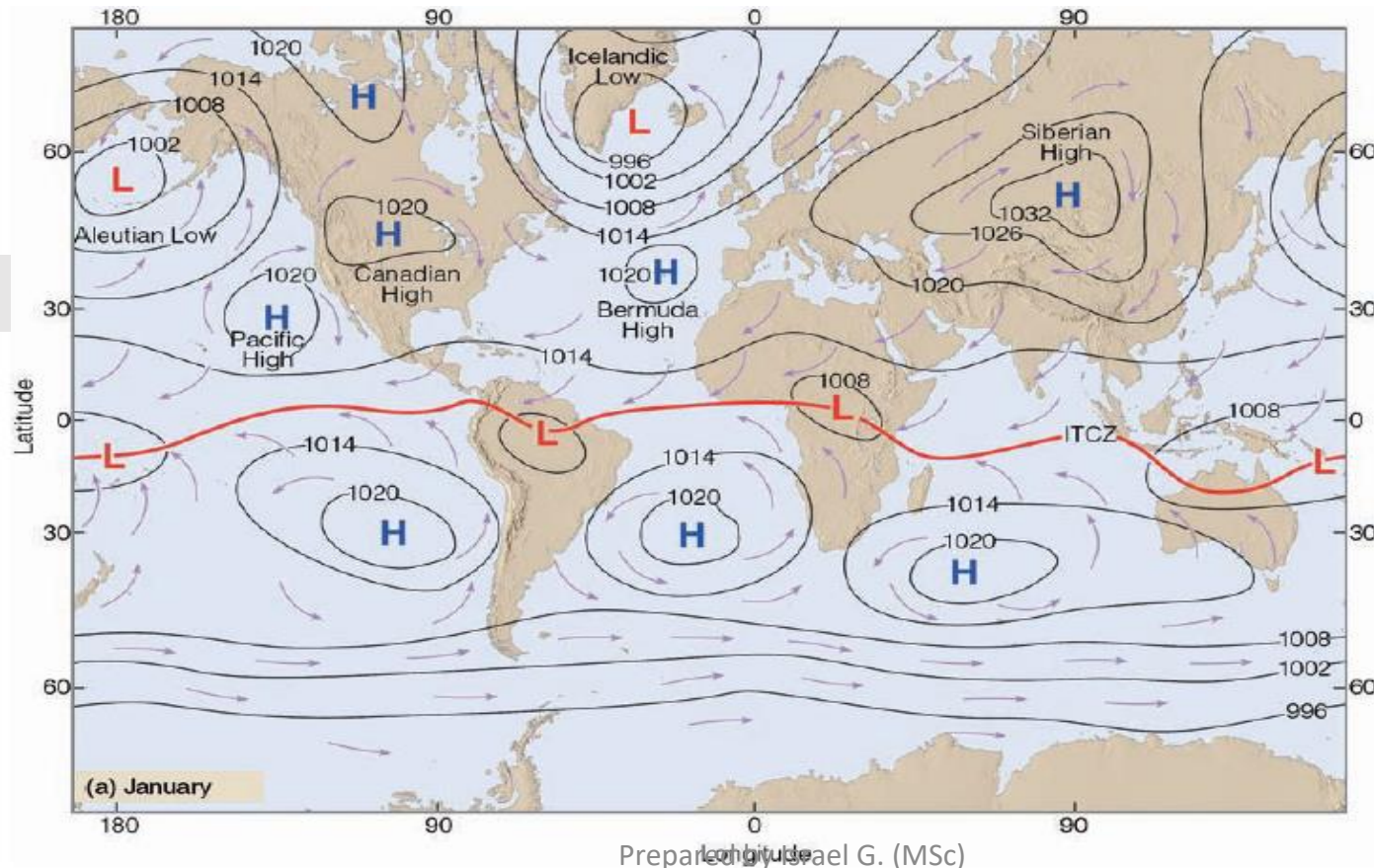


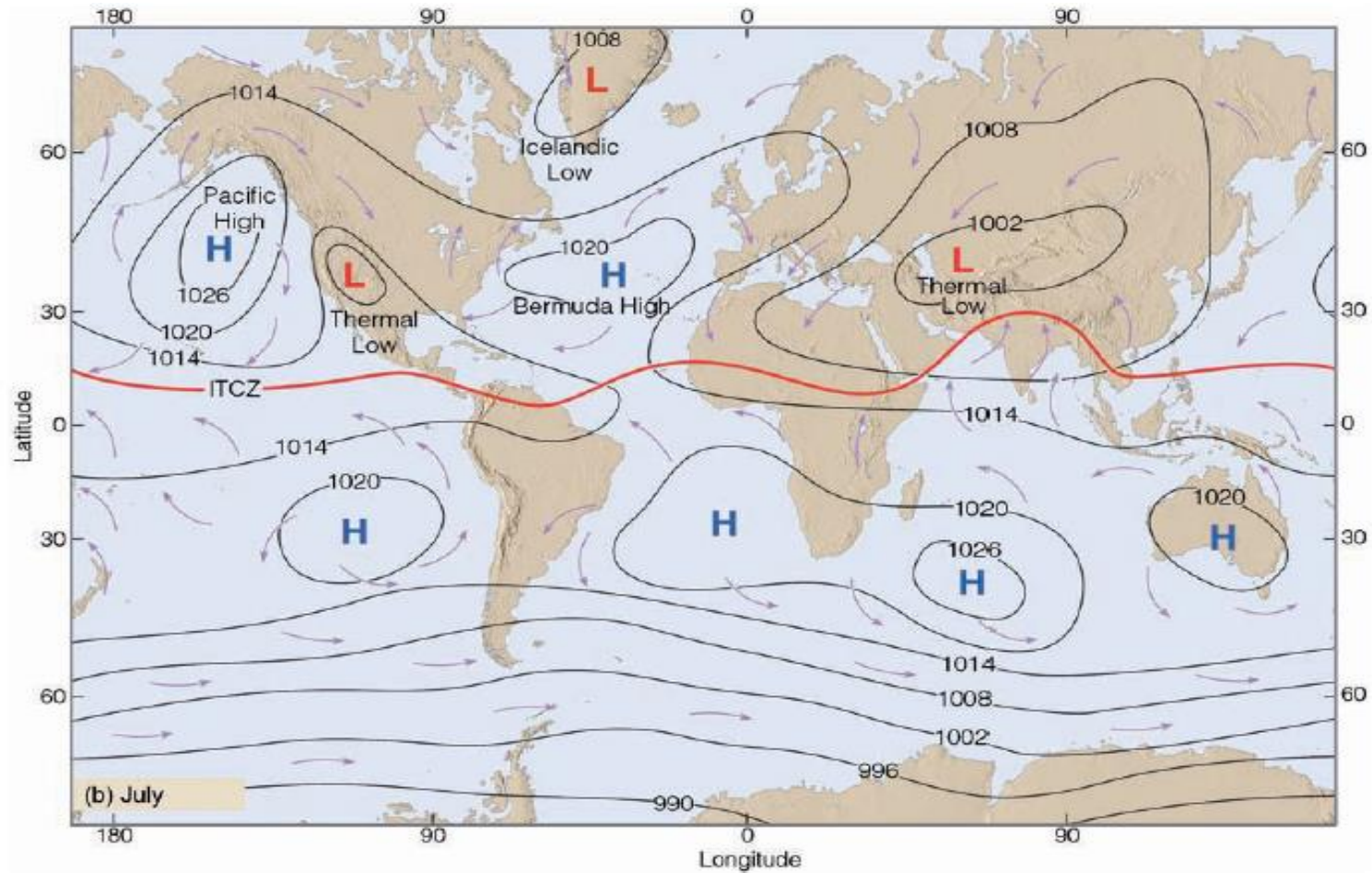
- ✓ As a result of the convergence the surface air rises at 60°N/S
- ✓ Some of the rising air returns at high levels to the middle latitudes, where it sinks back to the surface again.
- ✓ Some reaches the poles, slowly sinks to the surface and flows back to polar front

Average Surface Winds and Pressure: The Real World

- ✓ when we examine the real world with its continents and oceans, mountains and ice fields, we obtain an average distribution of sea-level pressure and winds for January and July, as shown in Fig a. and b.

January (a)





July (b)

The ozone layer

- ❑ The ozone layer is a region of Earth's stratosphere where the concentration of ozone (O₃) molecules is *significantly higher than* in the surrounding atmosphere.
- ❑ The ozone layer plays a crucial role in protecting life on Earth by absorbing the majority of the sun's harmful ultraviolet (UV) radiation.
- ❑ The ozone layer is composed of ozone molecules, which consist of three oxygen atoms (O₃)

- ❑ These high-energy rays can cause damage to living organisms, *including skin cancer, cataracts, and harm to ecosystems.*
- ❑ The formation and maintenance of the ozone layer involve a dynamic process known as the *ozone-oxygen cycle.*
- ❑ This cycle involves the dissociation and recombination of oxygen molecules (O_2) into ozone molecules (O_3) in the stratosphere.

- ❖ Human activities, particularly the release of *ozone-depleting substances* like chlorofluorocarbons (CFCs), have led to the thinning of the ozone layer, especially over polar regions. This phenomenon is commonly referred to as ozone depletion.
- ❖ Ozone holes, regions of significantly reduced ozone concentrations, have been observed primarily over Antarctica.

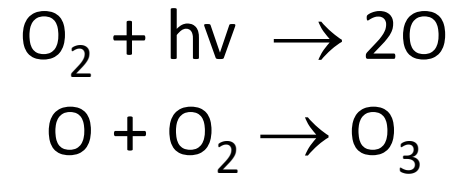
- These holes are associated with human-made substances that release chlorine and bromine into the stratosphere, contributing to ozone depletion.

Biogeochemical reactions in the atmosphere

- ✓ Biogeochemical reactions in the atmosphere involve chemical transformations of gases (often influenced by biological and physical processes) that regulate Earth's climate, air quality, and life-supporting systems.
- ✓ One important example is the ozone–oxygen cycle, which protects life from harmful ultraviolet (UV) radiation.
- ✓ Ozone is produced in the stratosphere by the disassociation of oxygen atoms that are exposed to shortwave solar radiation.

Ozone Formation (in the Stratosphere)

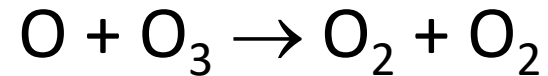
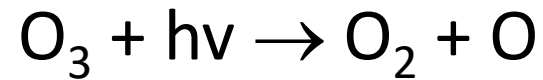
❑ Ozone is created naturally in the stratosphere by the absorption of ultraviolet light:



❑ Here, solar UV radiation breaks oxygen molecules (O_2) into individual oxygen atoms (O), which then combine with other O_2 molecules to form ozone (O_3).

Ozone Destruction by UV Radiation

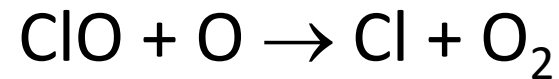
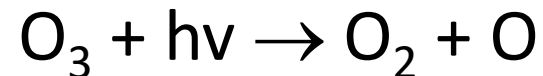
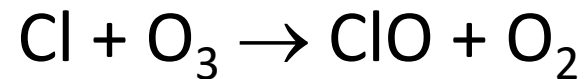
- ✓ Absorption of ultraviolet light at wavelengths between 200 and 320 nm destroys ozone:



- ✓ This is a natural photochemical balance ozone is continuously formed and destroyed, maintaining a steady concentration in the stratosphere.

Catalytic Destruction by Pollutants (Chlorine Cycle)

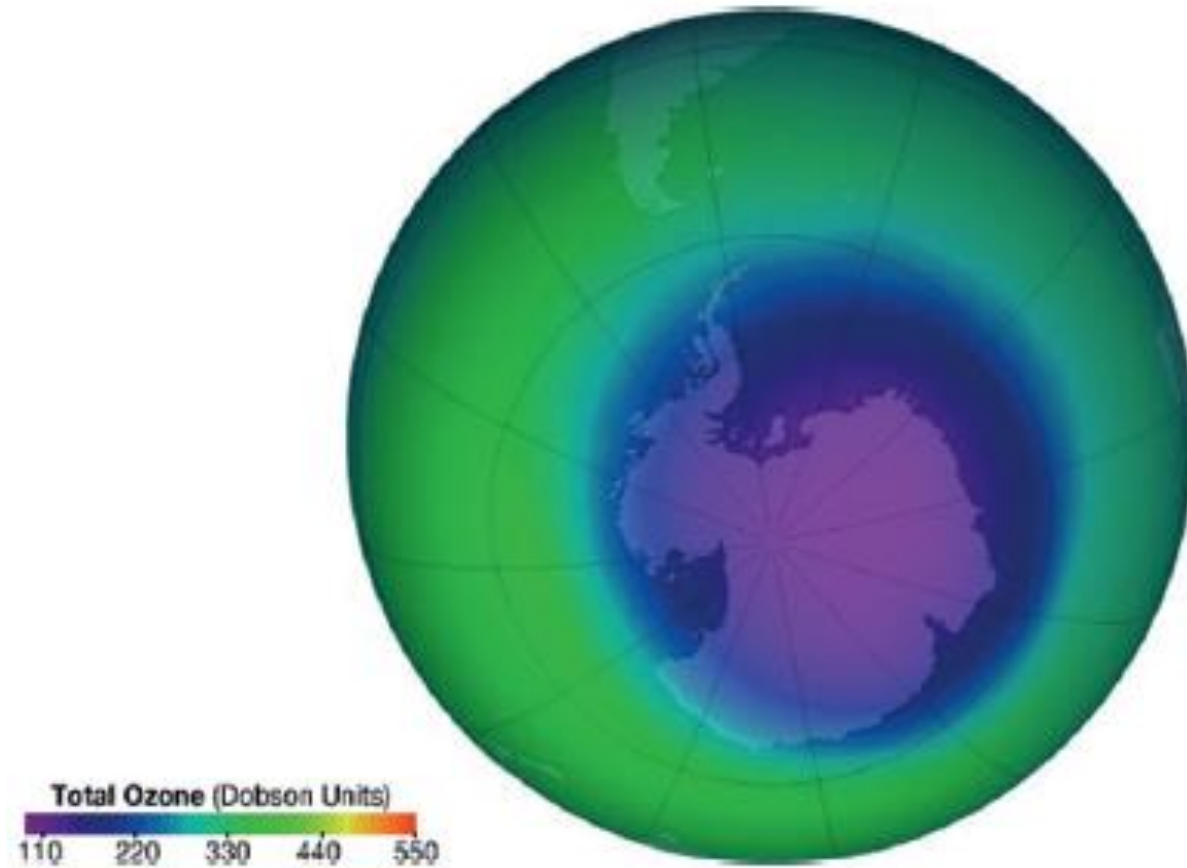
- ✓ Human-made chemicals, like chlorofluorocarbons (CFCs), release chlorine atoms (Cl) in the stratosphere, which catalytically destroy ozone:
- ✓ Stratospheric ozone is destroyed by chlorine, which acts as a catalyst in the reaction



- ✓ The chlorine atom is regenerated, so one Cl atom can destroy thousands of ozone molecules.
- ✓ For a net reaction of



- ✓ Since the mid-1980s, field measurements have indicated that the total density of ozone molecules in the atmospheric column has *declined* significantly globally.



The average abundance of ozone in the atmosphere of the Southern Hemisphere during October 2006. The ozone “hole,” seen in blue and purple, is actually an area where the abundance of ozone in the stratosphere is reduced—perhaps better described as a thinning rather than a hole.

Ozone Depletion vs. Ozone Hole

Ozone Depletion

❖ Ozone depletion refers to the gradual thinning or reduction of the concentration of ozone (O₃) in the stratosphere over time, *anywhere in the world*.

Cause:

- ✓ Mainly caused by human-made chemicals like chlorofluorocarbons (CFCs), halons, and other ozone-depleting substances (ODS).
- ✓ When these substances reach the stratosphere, UV light breaks them down, releasing chlorine (Cl) and bromine (Br) atoms that destroy ozone molecules:

Effect:

- ✓ Global reduction in ozone concentration.
- ✓ Leads to more UV radiation reaching Earth's surface, increasing risks of skin cancer, eye damage, and ecosystem harm.

Ozone Hole

❖ The ozone hole refers to a region of extremely low ozone concentration (severe ozone depletion) that occurs seasonally over *Antarctica* during spring (September–October).

Cause:

- ✓ Same process as ozone depletion, but strongly enhanced by special conditions in the Antarctic.

Effect:

- ✓ Dramatic drop in ozone levels up to 60–70% loss in the lower stratosphere.
- ✓ The “hole” is not a physical hole but an area where ozone concentration is much lower than normal.

Weather Patterns in the Atmosphere

- Weather is the day-to-day state of the atmosphere in a particular place.
- It is influenced by various factors, including temperature, humidity, air pressure, and wind.

Air Pressure and Wind:

- ❑ High and low-pressure systems play a crucial role in shaping weather patterns.
- ❑ The air moves from areas of high pressure to low pressure, creating wind.

Temperature and Atmospheric Stability:

- ❑ Temperature variations influence the stability of the atmosphere.
- ❑ Warm air tends to rise, creating areas of low pressure, while cold air sinks, creating high-pressure zones.

Humidity and Cloud Formation:

- ❑ Humidity, the amount of water vapor in the air, is a key factor in weather patterns.

Earth System Science

Chapter 2: THE COMPONENTS OF THE EARTH SYSTEM

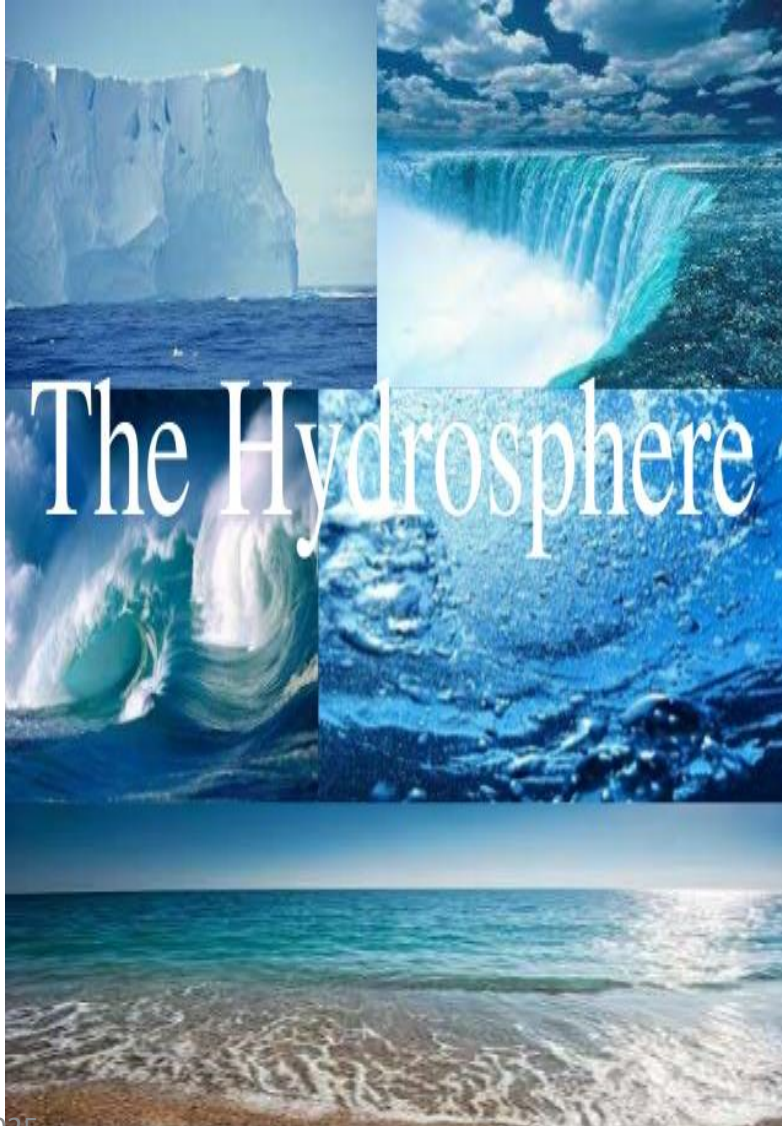


Objective

- *To Describe the Hydrosphere*
- *To Know The Ocean structure and circulation*
 - ✓ *Neutral Condition*
 - ✓ *El Niño Condition*
 - ✓ *La Niña Condition*
- *To Explain the composition of sea water*

The Hydrosphere

The Hydrosphere



- ✓ Water is the most abundant substance at the surface of the Earth.
- ✓ About 1.4 billion cubic km of water in liquid and frozen form make up the oceans, lakes, streams, glaciers, and ground waters.
- ✓ this enormous volume of water enclosing much of the terrestrial surface is known as the hydrosphere.

- ❖ Hydrosphere is *the total amount of water on, under, and above the Earth's surface* including oceans, rivers, lakes, glaciers, groundwater, and water vapor in the atmosphere.
- ❖ hydrosphere interacts with the atmosphere, geosphere, biosphere, and anthroposphere through energy and matter exchange (mainly via the water cycle).

Distribution and Quantity

- Oceans are the largest stock of earth's water, 96.5%
- 1.7% is in polar ice
- 1.7% is in ground water
- Only 0.1% is in the surface and atmospheric water systems
- Only 0.0002% of earth's water is contained in rivers

Looking at fresh water only:

- Polar ice accounts the largest share, 68.6%
- Fresh ground water account 30.1% of earth's fresh water
- Fresh lakes constitute 0.26% of fresh water
- Only 0.006% of fresh water is contained in rivers

Estimated world water quantity

Location		Area (10 ⁶ km ²)	Volume (km ³)	Percent of total water	Percent of fresh water
Oceans		361.3	1,338,000,000	96.5	
Ground water	Fresh	134.8	10,530,000	0.76	30.1
	Saline	134.8	12,870,000	0.93	
Soil moisture		82	16,500	0.0012	0.05
Polar ice		16	24,023,500	1.7	68.6
Other ice and snow		0.3	340,600	0.025	1
Lakes	Fresh	1.2	91,000	0.007	0.26
	Saline	0.8	85,400	0.006	
Marshes		2.7	11,470	0.0008	0.03
Rivers		148.8	2,120	0.0002	0.006
Biological water		510	1,120	0.0001	0.003
Atmospheric water		510	12,900	0.001	0.04
Total water		510	1,385,984,610	100	
Fresh water		148.8	35,029,210	2.5	100

Global annual water balance

- Though the water content of surface and the atmosphere is small, immense quantity of water annually pass through them

Flux		Ocean (Area = $361.3 \times 10^6 \text{ km}^2$)	Land (Area = $148.8 \times 10^6 \text{ km}^2$)
Precipitation		458,000 (km^3/yr)	119,000 (km^3/yr)
Evaporation		505,000 (km^3/yr)	72,000 (km^3/yr)
Runoff to ocean	Rivers	-	44,700 (km^3/yr)
	Ground water	-	2200 (km^3/yr)
	Total runoff	-	47,000 (km^3/yr)

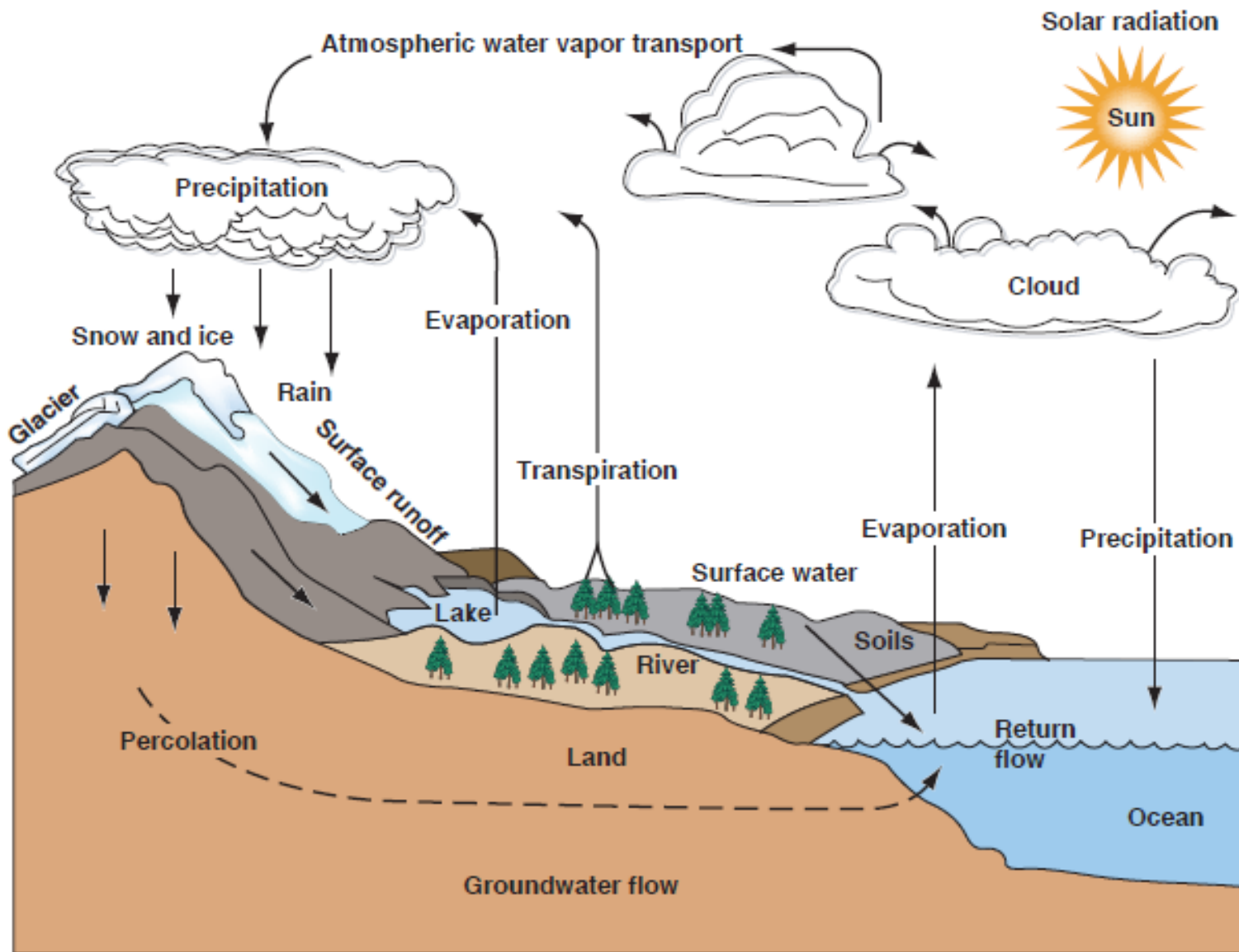
- Evaporation from land consumes 61% of land precipitation, where will the 39% go? Forms runoff to oceans as surface water

The Hydrologic Cycle

The Hydrologic cycle (the water cycle)

- This refers to the routine circulation of water through network of paths in the hydrosphere
- Water evaporates from oceans and land surface to become atmospheric moisture.
- The moisture is transported and lifted until it condenses and then precipitate.
- Precipitated water may be intercepted by vegetation; may become overland flow; may discharge into stream as surface runoff

- Or infiltrate into the ground and flow as subsurface flow
- Or the infiltrated water may percolate deeper recharge the ground water
- It may later emerge in springs or seeping into streams
- Finally reaching the sea or evaporating into the atmosphere



The Hydrologic Cycle

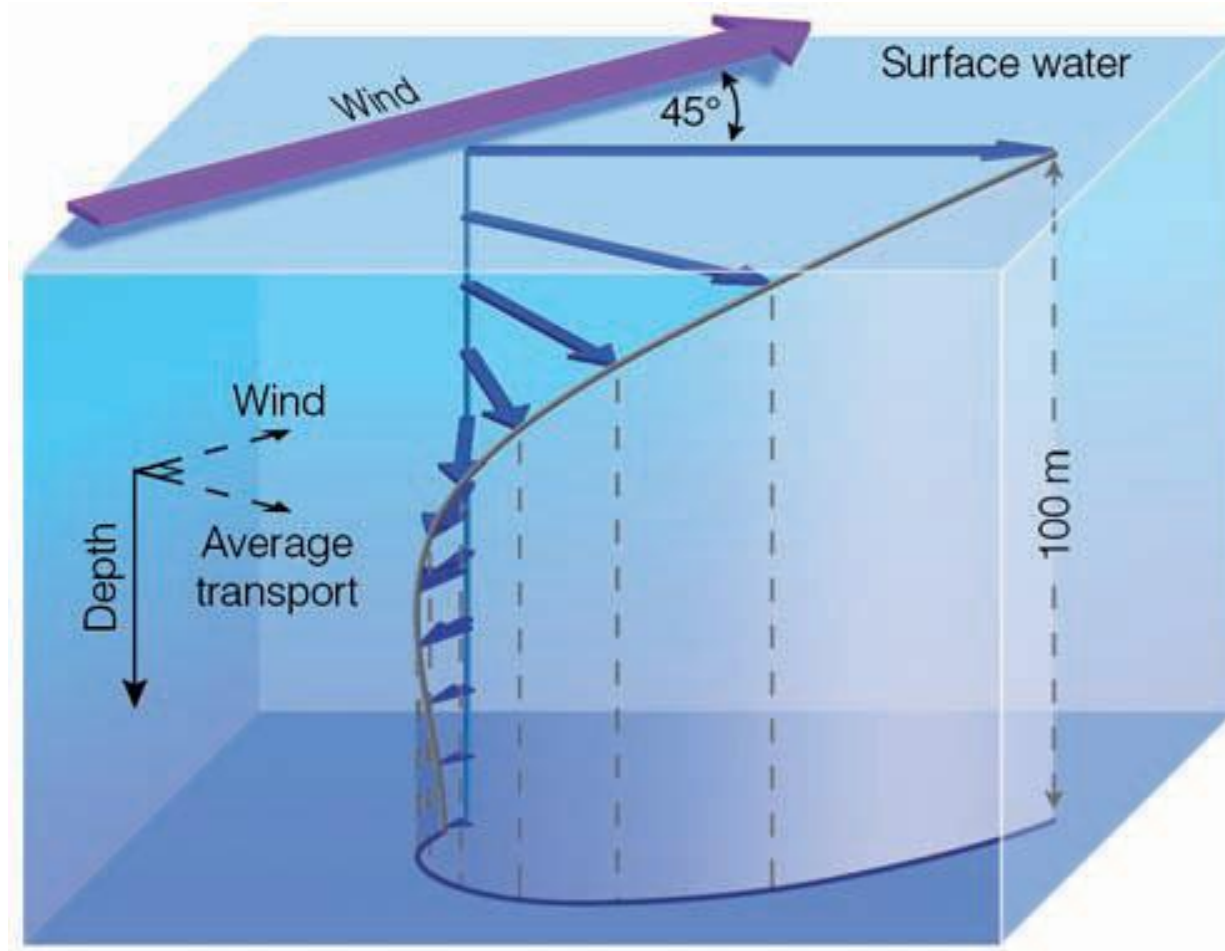
The Global Water (Hydrologic) Cycle

- ❖ Explain how water circulates continuously among the hydrosphere's reservoirs through:
 - ✓ Evaporation (from oceans, lakes, soils)
 - ✓ Transpiration (from plants)
 - ✓ Condensation (cloud formation)
 - ✓ Precipitation (rain, snow)
 - ✓ Infiltration and percolation (into soil and groundwater)
 - ✓ Runoff (back to oceans and rivers)

The Ocean Structure And Circulation

The Ocean structure and circulation

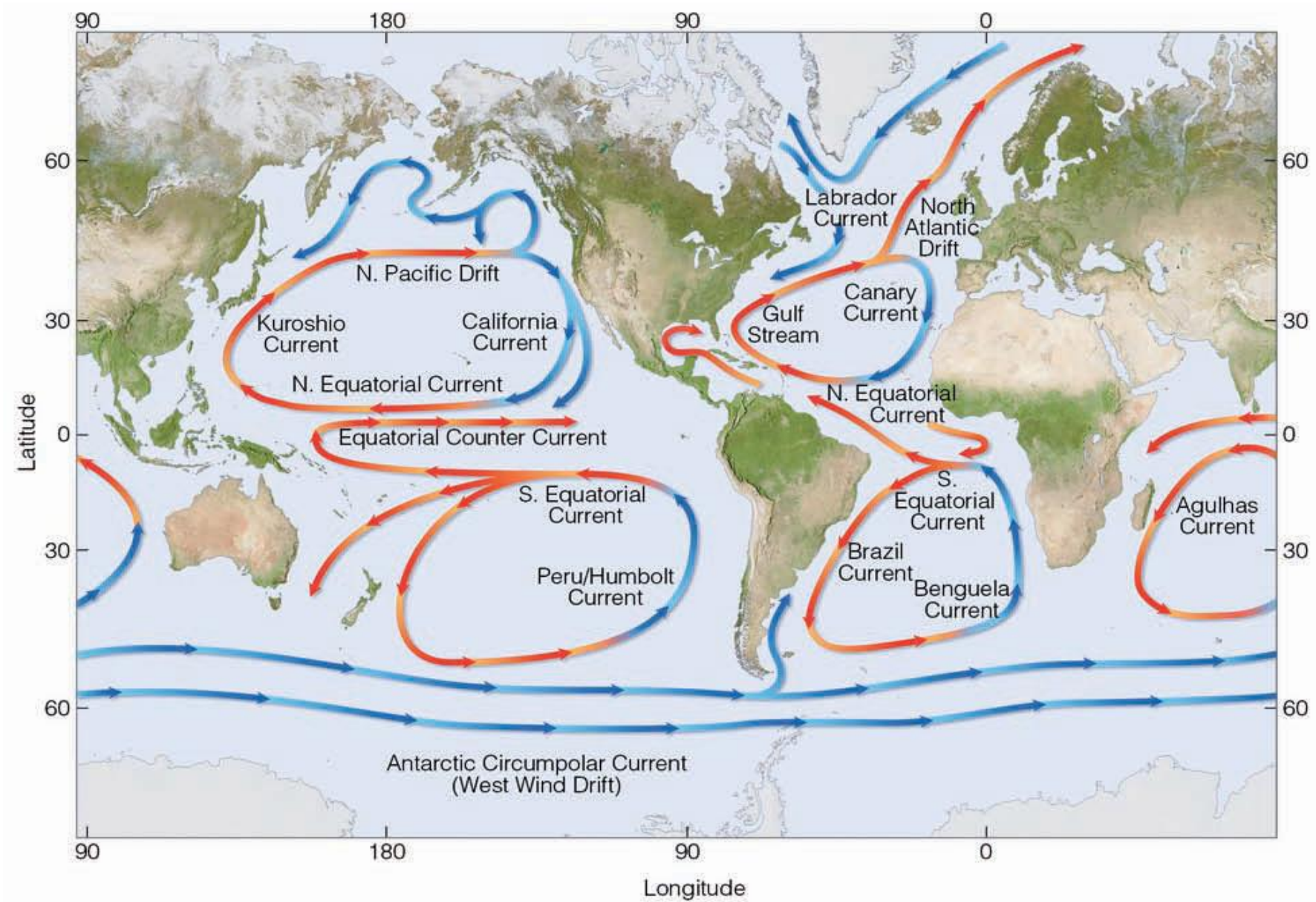
- As the wind blows over the oceans, it causes the surface water to drift along with it.
- The Coriolis force bends the moving water to the right in the Northern Hemisphere.
- If we look at a shallow surface layer of water, we see that it moves at an average angle of about 45° to the direction of the wind.
- Below the surface each successive layer of water moves more slowly and is deflected to the right of the layer above.



The Ekman Spiral. As the wind blows over an open stretch of ocean, the surface water beneath it is set in motion, and the Coriolis force deflects the water to the right in NH

- The moving water gradually piles up, creating pressure differences within the water itself.
- This leads to further motion several hundreds of meters down into the water.
- Surface water tends to move in a circular pattern as winds blow outward, away from the center of the subtropical highs.
- These circular whirls of ocean currents are called *gyres*

- Generally, west coasts are dominated by warm currents and east coasts are dominated by cold currents.
- Warm currents are often directed poleward whereas cold currents are directed to equatorward.



The El Niño-Southern Oscillation (ENSO)

- The El Niño-Southern Oscillation (ENSO) describes the natural year-to-year variations in the ocean and atmosphere in the tropical Pacific.
- Sea-surface temperatures in the central and eastern equatorial Pacific cycle between above-average and below-average.
- *An El Niño state* occurs when the **central and eastern equatorial Pacific sea-surface temperatures** are substantially **warmer** than usual.
- Because this condition frequently occurs around Christmas, Peruvian fishermen called it *El Niño* which means a boy child in Spanish.

- *El Niño* refers to the ocean component of ENSO.
- The *Southern Oscillation* part of the term ENSO refers to the atmospheric component: which refers to the **fluctuation of atmospheric pressure** between the central/eastern Pacific and the western Pacific.
- La Niña conditions occur when the central and eastern equatorial Pacific waters are substantially cooler than usual.

What are neutral conditions?

- In order to understand the ENSO cycle, we must first understand what the Pacific is like in its neutral state.
- The equatorial Pacific climate acts as a “*coupled* system” because the state of the ocean and atmosphere depend on each other.
- As the conditions of the ocean change, the atmosphere responds, and vice versa.
- The main indicators of these changes are **pressure** and **temperature**.

- Normally, in the tropical Pacific Ocean, the trades are persistent winds that blow **westward**.
- As trade winds blow over the open stretch of the equatorial pacific ocean, the surface water beneath them moves from **east to west**.
- As this water moves westward, it is heated by sunlight and the atmosphere.
- When the surface water arrives in the western Pacific, land surfaces restrict its further movement and so the water “piles up”: As the water accumulates and stagnates, it continues to warm.
- So, the trades create **upwelling in the eastern pacific** that brings cold water to the surface.

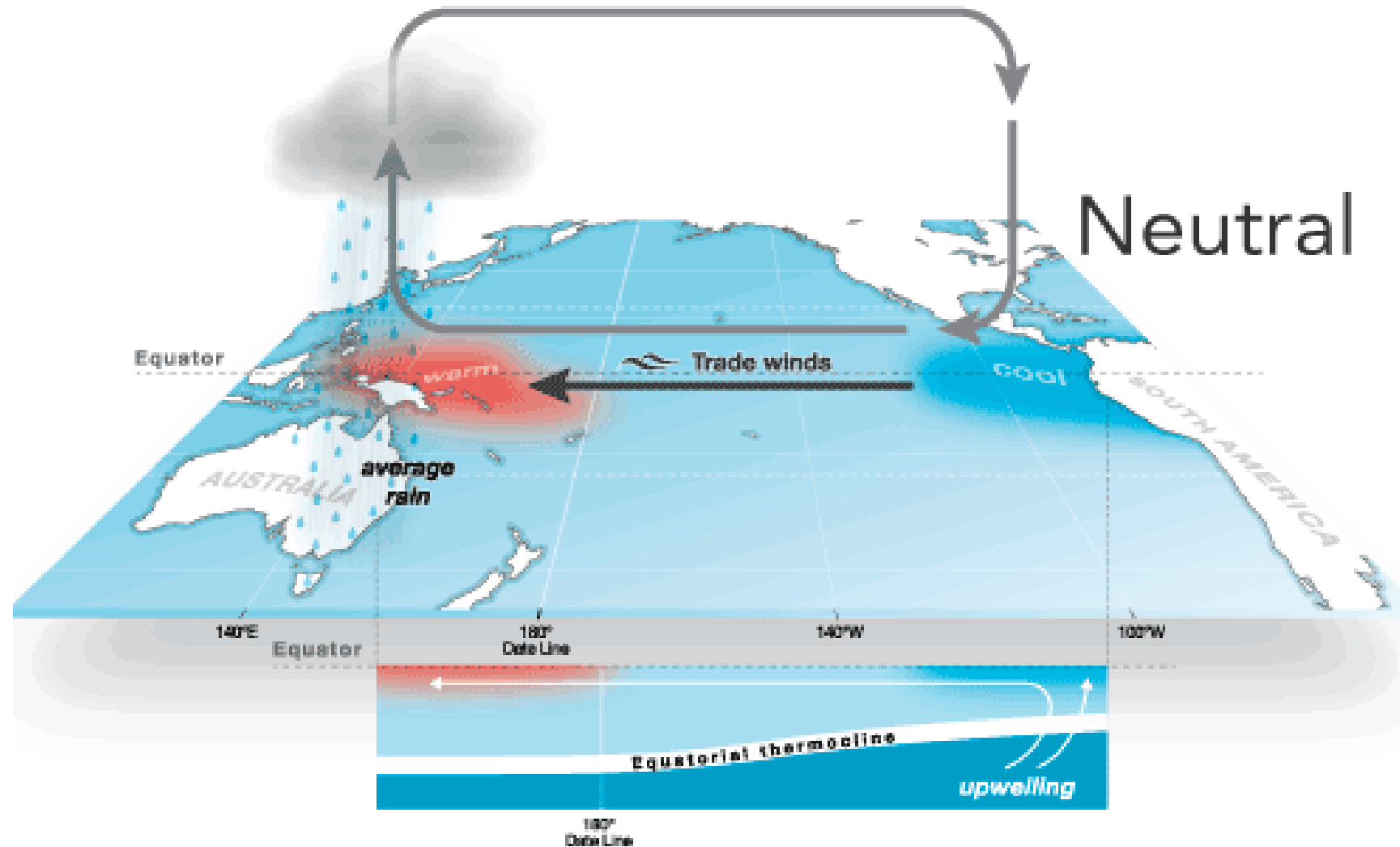
Ocean Temperature?

- During neutral conditions, waters in the eastern Pacific are upwelled along the equator and the coast of South America.
- Because the upwelled equatorial and coastal waters originate from deep below the surface, they are **cold** and **rich with nutrients**.
- Where as **western equatorial** pacific water is **warm**

Air Pressure?

- During neutral conditions atmospheric pressure is **low in the western** tropical Pacific (referenced at Darwin, Australia), and relatively **higher in the cooler central/eastern** tropical Pacific (referenced at Tahiti).

Neutral Condition

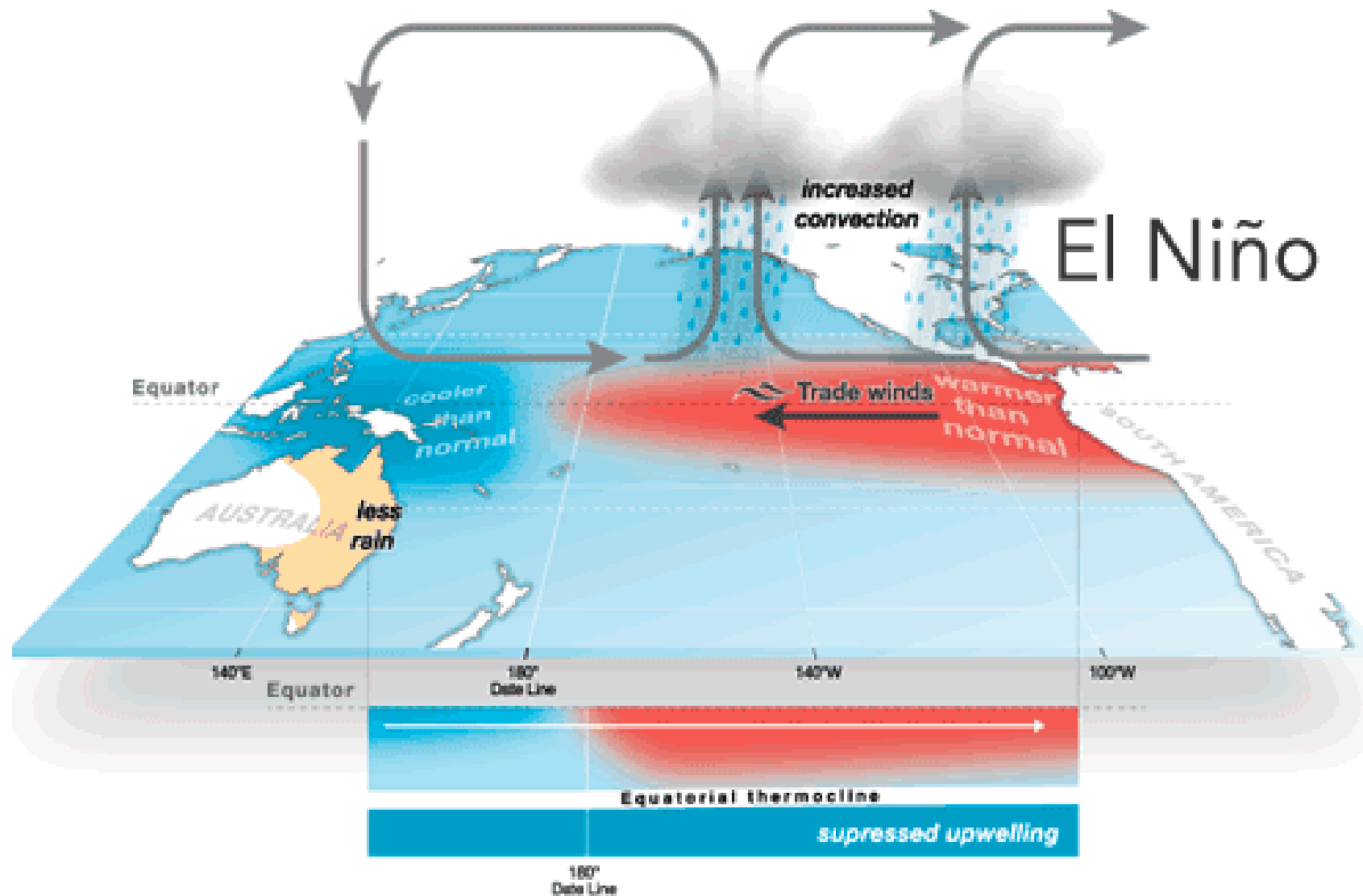


Normal Pacific Condition

What happens during *El Niño* (Warm Phase)?

- The start of El Niño is initiated when trade winds are weekend or sometimes are reversed
- This is because every few years, the surface atmospheric pressure patterns break down, as air pressure rises over the region of the western Pacific and falls over the eastern Pacific.
- Surface water warms over a broad area of the tropical Pacific and heads eastward
- During strong El Niño condition, the easterly trade winds may become westerly winds.
- The upwelling is suppressed in the eastern pacific
- The convective activity moves eastward.

El Niño Condition



- Such a large area of abnormally warm water can have an *effect on global wind patterns*.
- The warm tropical water fuels the atmosphere with additional warmth and moisture, which the atmosphere turns into additional storminess and rainfall.
- The added warmth from the oceans and the release of latent heat during condensation apparently influence the westerly winds aloft in such a way that certain regions of the world experience too much rainfall, whereas others have too little.
- Over the warm tropical central Pacific, the frequency of *typhoons* usually increases.

- Although the actual mechanism by which changes in surface ocean temperatures influence global wind patterns is not fully understood, the by products are plain to see.
- During exceptionally warm El Niño, **drought** is normally felt in **Indonesia, southern Africa, and Australia**, while heavy rains and flooding often occur in **Ecuador and Peru**.
- This extremely warm episode, which occurs at irregular intervals of **two to seven years**

El Niño's Impact:

Weather:

- ✓ Disruption of normal weather patterns leads to floods, droughts, and extreme temperatures.

Marine Ecosystems:

- ✓ Decline in fish populations due to reduced upwelling.

Global Economy:

- ✓ Impacts agriculture, energy demand, and disaster management.

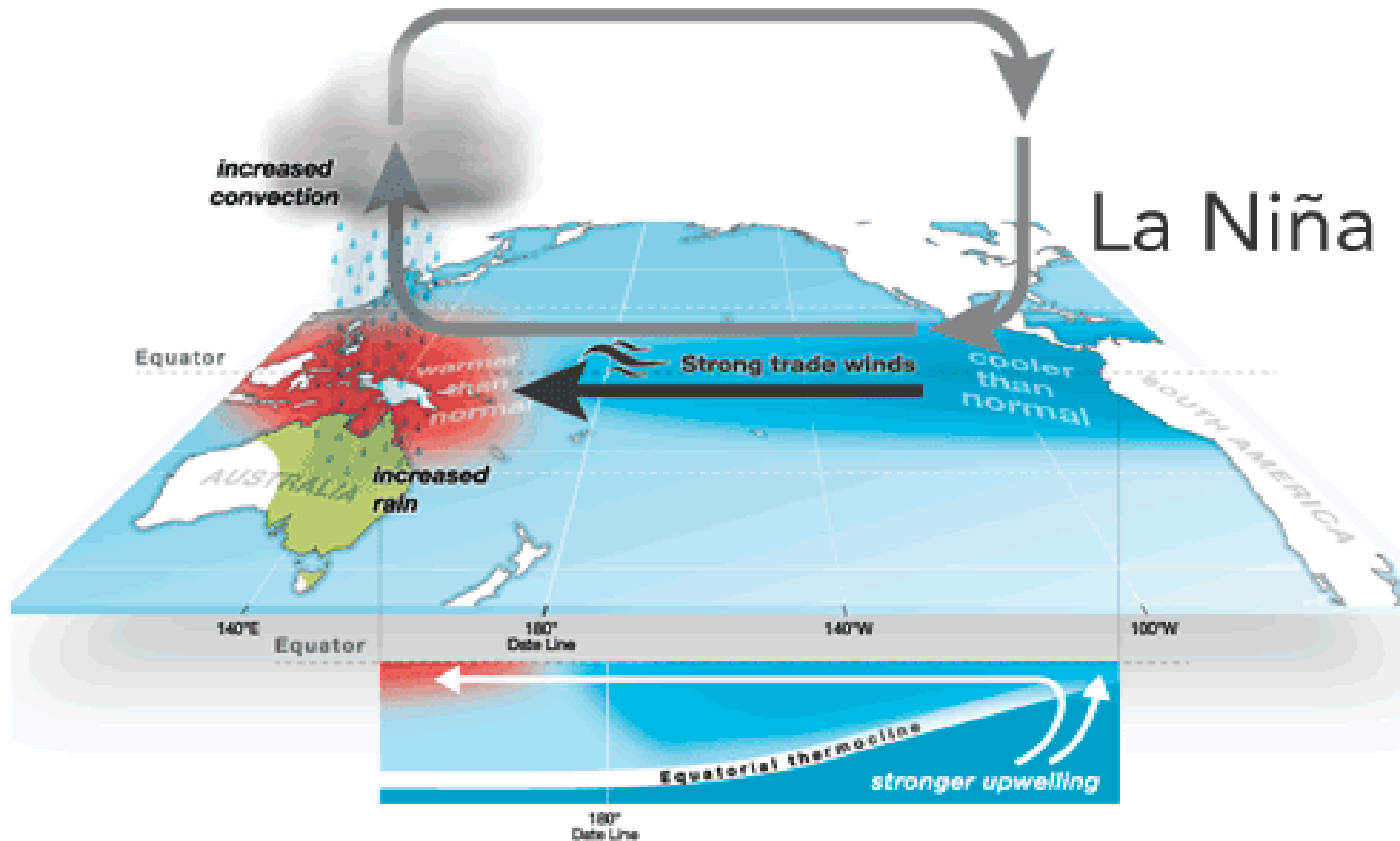
Human Health:

- ✓ Increases risks of disease outbreaks due to changing environments.

What happens during La Niña (*Cold Phase*)?

- La Niña (girl child) occurs as an enhanced version of the neutral state.
- Air pressure is higher-than-average in Tahiti (eastern Pacific) and lower-than-average in Australia (western Pacific)

- The easterly trade winds blow more intensely than usual.
- The colder-than-normal ocean water extends across the eastern and central equatorial Pacific
- Like with the neutral condition, the convective activity is in the western pacific, but with higher intensity



The Composition Of Sea Water

	Ocean Water	Sea Water
Definition	Refers specifically to water in the open ocean , which covers most of Earth's surface.	Refers to water found in smaller, often enclosed or semi-enclosed seas (e.g., Mediterranean Sea, Baltic Sea).
Salinity Variations	Salinity is relatively consistent at 35 ppt in open ocean water, except in areas near river outflows or polar regions.	Salinity varies more widely, ranging from 10–42 ppt , depending on the sea's location and size.
Freshwater Influence	Less influenced by rivers and freshwater sources due to the vast volume of the open ocean.	Often more influenced by rivers, evaporation, or precipitation, leading to larger salinity and chemical variability.
Nutrient Content	Ocean water typically has balanced nutrient levels, supporting diverse ecosystems.	Sea water can have varying nutrient levels, often higher near coastal areas due to river input or upwelling.
Temperature Range	More stable temperatures due to the large volume and depth of the ocean.	Larger temperature fluctuations due to smaller size and proximity to land.

- ✓ Seawater, or salt water, is water from a sea or ocean.
- ✓ On average, seawater in the world's oceans has a salinity of about 3.5% (35 g/L, 35 ppt, 600 mM).
- ✓ This means that every kilogram (roughly one liter by volume) of seawater has approximately 35 grams (1.2 oz) of dissolved salts (predominantly sodium (Na^+) and chloride (Cl^-) ions).
- ✓ Seawater contains more dissolved ions than all types of freshwater.

- ✓ The most abundant dissolved ions in seawater are **sodium, chloride, magnesium, sulfate and calcium.**
- ✓ Small amounts of other substances are found, including amino acids at concentrations of up to 2 micrograms of nitrogen atoms per liter, which are thought to have played a key role in the origin of life.



....END

Thanks !